

AlloQuant - output-file manifest

Reference run: CSK-SRC dimer scan, 260718 rerun (AF3-complete) | generated 2026-07-21

- AlloQuant is a two-module pipeline. Module 1 (structural extraction) writes the run-root files below; Module 2 (allostery discovery) writes one plots_and_stats_{KINASE}_{METHOD}/ folder per analysed kinase.
- Tokens: {KINASE} = analysed kinase (CSK, SRC, CDK1, ...); {METHOD} = clustering method (GMM default, or KMEANS); {ver} = pipeline version (v7r2); {i}/{j} = discovered-state indices.
- For this CSK-SRC rerun there are two Module-2 folders: plots_and_stats_CSK_GMM/ and plots_and_stats_SRC_GMM/ (identical file set, different target kinase). The parallel *_NO_PY159APO/ folders hold the same analysis with the SRC pY159-apo condition excluded (sensitivity check).
- Interactive HTMLs need their sibling *_files/ asset folder to render offline; keep the pair together when sharing.
- MAC = Mean Allosteric Coupling (global coupling score) = mean of absolute metric-metric correlation-edge weights; used both per-condition (Phase 5) and per-state (Phase 6b).
- 'exists (this run)' is auto-checked against plots_and_stats_CSK_GMM/ (Module 2) and the run root (Module 1).

#	Stage	Phase	File / pattern	Format	Description	exists (this run)
1	Module 1	Setup	experiment.csv	CSV	Experiment design table: one row per dimer condition (chain_a, condition_a, chain_b, ptm_b, condition_b). Defines the systems built and analysed.	yes
2	Module 1	Setup	generated_matrix.csv	CSV	Combinatorial multimer matrix (chain-A rows x chain-B columns) marking which pairwise AF3 jobs were generated.	yes
3	Module 1	Setup	fasta_source_map.json	JSON	Maps each construct label to its source FASTA sequence and provenance (used to identify/type each chain).	yes
4	Module 1	Landmarks	hmm_landmarks.json	JSON	Per-kinase-type HMM landmark residue indices (VAIK-Lys k, alphaC-Glu c, R-spine rs1/rs2, DFG f, HRD, and numbered allosteric landmarks). Consumed by every distance metric.	yes
5	Module 1	Master	master_kinase_analysis_results_{ver}.csv	CSV	PRIMARY OUTPUT: one row per protein chain per model, with all conformational labels and allosteric distances. Documented column-by-column in the companion data dictionary.	yes
6	Module 2	Phase 0	Phase0_AF3_Confidence_Metrics.pdf	PDF	AF3 confidence & QC plots (ipTM / pTM / PAE distributions) across conditions.	yes
7	Module 2	Phase 0	Phase0_AF3_Quality_Summary.csv	CSV	Per-condition AF3 confidence/quality summary table underlying the Phase 0 plots.	yes
8	Module 2	Phase 1	Phase1_AF3_Ligand_Placement_Preference.pdf	PDF	System-level ligand occupancy ('tug-of-war'): where AF3 places ATP/ligand between the two chains. Ligand-bearing systems only (absent for ligand-free targets).	yes
9	Module 2	Phase 1	Phase1_Ligand_QC_Review.csv	CSV	Per-model ligand-placement QC flags and mismatch-override decisions (apo/holo assignment).	yes
10	Module 2	Phase 2	Phase2_Macro_States.pdf	PDF	Categorical macro-state composition (State labels) per condition.	yes
11	Module 2	Phase 2	Phase2_Pairwise_Categorical_Stats.csv	CSV	Pairwise categorical tests (state-label distributions) between conditions.	yes
12	Module 2	Phase 3	Phase3_2D_PhaseSpace.pdf	PDF	2D conformational phase space: DFG dihedral (Phi/Psi) and Dunbrack spatial (D1/D2) density panels, coloured by condition.	yes
13	Module 2	Phase 4	Phase4_Allosteric_Distances.pdf	PDF	1D distributions (violins) of every allosteric-network distance metric across conditions.	yes
14	Module 2	Phase 4	Phase4_Kruskal_Summary.csv	CSV	Kruskal-Wallis omnibus test per distance metric across conditions.	yes
15	Module 2	Phase 4	Phase4_Pairwise_Summary.csv	CSV	Post-hoc pairwise comparisons (with effect sizes) per metric between conditions.	yes
16	Module 2	Phase 5	Phase5_Correlation_Heatmaps.pdf	PDF	Per-condition metric-metric correlation heatmaps (the coupling networks).	yes
17	Module 2	Phase 5	Phase5_Correlation_Shift_Volcanos.pdf	PDF	Volcano plots of correlation-edge shifts between conditions (which couplings strengthen/weaken).	yes
18	Module 2	Phase 5	Phase5_Differential_Correlations.csv	CSV	Per-edge differential-correlation table between conditions (values behind the shift volcanos).	yes
19	Module 2	Phase 5	Phase5_Global_Network_Density.csv	CSV	Global coupling score (MAC = mean absolute correlation-edge weight) per condition.	yes
20	Module 2	Phase 5	Phase5_Global_Network_Density.pdf	PDF	Plot of the per-condition global coupling score (network density).	yes
21	Module 2	Phase 5	Phase5_Global_Network_Density_Stats.csv	CSV	Pairwise statistics comparing global coupling scores between conditions.	yes
22	Module 2	Phase 5	Phase5_PCA_Biplot.pdf	PDF	PCA biplot of conditions in the allosteric-metric space (loadings + scores).	yes
23	Module 2	Phase 5	Phase5_Interactive_3D_Condition_Space.html	HTML	Interactive 3D (plotly) view of conditions in metric space. Companion assets live in Phase5_Interactive_3D_Condition_Space_files/.	yes
24	Module 2	Phase 5	Phase5_Interactive_3D_Condition_Space_files/	dir (assets)	Supporting JS/CSS libraries for the Phase 5 interactive HTML (plotly, htmlwidgets, jquery). Ship alongside the .html.	yes
25	Module 2	Phase 6	Phase6_State_Assignments.csv	CSV	Unsupervised (GMM/KMEANS) state-cluster assignment per model (State 1..N).	yes
26	Module 2	Phase 6	Phase6_State_Clusters_PCA.pdf	PDF	PCA scatter of models coloured by discovered state cluster.	yes
27	Module 2	Phase 6	Phase6_State_Composition.pdf	PDF	Composition of each discovered state by condition (how conditions populate the states).	yes
28	Module 2	Phase 6	Phase6_Interactive_3D_State_Space.html	HTML	Interactive 3D (plotly) view of models coloured by state. Assets in Phase6_Interactive_3D_State_Space_files/.	yes
29	Module 2	Phase 6	Phase6_Interactive_3D_State_Space_files/	dir (assets)	Supporting JS/CSS libraries for the Phase 6 interactive HTML.	yes

#	Stage	Phase	File / pattern	Format	Description	exists (this run)
30	Module 2	Phase 6b	Phase6b_State_Intrinsic_Rigidity_MAC.csv	CSV	Per-state intrinsic mechanical rigidity: within-state MAC (mean absolute pairwise coupling); higher = more rigid/coupled.	yes
31	Module 2	Phase 6b	Phase6b_State_Intrinsic_Rigidity.pdf	PDF	Plot of per-state intrinsic rigidity (MAC).	yes
32	Module 2	Phase 6b	Phase6b_State_Intrinsic_Rigidity_Stats.csv	CSV	Pairwise Wilcoxon statistics on per-state intrinsic MAC (source of manuscript Supplementary Table 1).	yes
33	Module 2	Phase 7	Phase7_Complete_Structural_Metadata.csv	CSV	Full merged per-model metadata: master metrics plus derived condition/state columns (analysis-ready table).	yes
34	Module 2	Phase 7	Phase7_MacroState_Signatures.pdf	PDF	Biological-signature profiles mapping discovered states to structural features.	yes
35	Module 2	Phase 7	Phase7_MacroState_Signatures_Stats.csv	CSV	Statistics behind the macro-state signature profiles.	yes
36	Module 2	Phase 8	Phase8_Volcanos/	dir	Differential allosteric-driver analysis: one volcano per ordered pair of discovered states. For N states = C(N,2) pairs; here 9 states -> 36 pairs.	yes
37	Module 2	Phase 8	Phase8_Volcanos/Stats_State_{i}_vs_State_{j}.csv	CSV	Per-metric differential statistics (signed effect size, -log10 p) for state i vs state j.	(pattern)
38	Module 2	Phase 8	Phase8_Volcanos/Volcano_State_{i}_vs_State_{j}.pdf	PDF	Volcano plot of the metrics that most distinguish state i from state j.	(pattern)